

KELVIN WU

604-832-6169 | kelvinwu0002@gmail.com | github.com/taco58

EDUCATION

University of British Columbia

Vancouver, BC

Bachelor of Science – Combined Major in Computer Science and Statistics | **GPA: 92.5%**

Expected May 2029

Relevant Coursework: Data Structures & Algorithms, Software Construction & OOP, Models of Computation, Systematic Program Design

PROJECTS

Beam – Peer-to-Peer Network File Transfer CLI | GitHub

Sep 2026

Python, TCP/UDP, Sockets, Threading, HTTP, Click, Rich, Pytest

- Built a peer-to-peer file transfer CLI using **UDP broadcast** for automatic device discovery and **TCP** for reliable transfers, eliminating manual IP configuration on local networks.
- Designed a custom binary protocol that streams files in **64 KB chunks**, keeping memory usage below **1 MB** regardless of file size.
- Added **SHA-256 integrity checks**, atomic **.part** files, filename collision handling, and interrupted-transfer recovery to prevent corrupted or partially written downloads.
- Implemented an embedded **HTTP server** with terminal-generated QR codes, allowing phones and other devices to download shared files without installing the CLI.

Distributed Uptime Monitoring Platform | GitHub

Aug 2026

Go, Redis, PostgreSQL, Next.js, Docker, Kubernetes, Helm, Terraform, AWS, GitHub Actions

- Built a distributed monitoring service in **Go** that runs health checks across Kubernetes replicas, using **Redis distributed locks** to ensure each check executes only once.
- Stored monitor configurations and check results in **PostgreSQL**, with queries for **24-hour uptime** and **P95 latency** and automated cleanup of stale results to keep historical data manageable.
- Built a **JWT-authenticated REST API** with Discord webhook alerts for monitor failures and recoveries, allowing users to respond to incidents without continuously checking the dashboard.
- Automated deployment to **AWS EKS** with Terraform, Helm, and GitHub Actions, replacing manual infrastructure and deployment steps with a repeatable CI/CD pipeline.

Adaptive TDEE & Health Analytics Platform | GitHub

Jun 2026

Next.js, TypeScript, React, Tailwind CSS, Supabase, PostgreSQL

- Built a full-stack analytics platform used by **15+ daily active users** with **1,000+ logged entries**, using an EMA-based algorithm to reduce day-to-day noise in weight and calorie trends.
- Implemented Supabase PostgreSQL Row-Level Security and authenticated Next.js Server Actions to isolate user data at the database layer rather than relying only on frontend checks.
- Built responsive React/Tailwind dashboards that turn logged calorie and weight data into historical trends for tracking estimated expenditure and progress.

EXPERIENCE

Competitive Programming Instructor

Remote

Private Tutor

Sep 2024 – Apr 2025

- Taught **40+ hours of C++** to 15+ students, covering data structures, algorithms, recursion, BFS/DFS, dynamic programming, and Big-O analysis for the CCC Junior contest.
- Ran live debugging sessions where students traced **algorithmic, logical, and memory-related bugs**, developing systematic approaches to breaking down and optimizing problems.

Kitchen Staff

White Rock, BC

Whitecliff Senior Retirement Home

Jul 2024 – Aug 2025

- Supported meal service for **150+ residents** across three daily dining periods while coordinating with culinary staff in a fast-paced, time-sensitive environment.

TECHNICAL SKILLS

Languages: Go, Python, C++, TypeScript, JavaScript, SQL, Java

Frameworks & Libraries: Next.js, React, Tailwind CSS, Supabase, Click, Rich, Pytest

Databases & Cloud/DevOps: PostgreSQL, Redis, Docker, Kubernetes, Helm, Terraform, AWS (EKS, RDS, ECR), GitHub Actions

Networking & APIs: TCP/UDP Sockets, REST, JWT, HTTP, WebSockets